

Handling Features with More Than Two Categories

The Problem

So far, our features have been binary (e.g., "Pointy" vs. "Floppy"). What happens if a feature has *more than two* discrete values?

- **Example:** Ear Shape can now be "Pointy", "Floppy", or "Oval".

While you *could* build a tree that splits into three branches, a more common and robust method is to use **one-hot encoding**.

The Solution: One-Hot Encoding encode

One-hot encoding is a technique that transforms a single categorical feature into multiple new **binary (0 or 1) features**.

- **Rule:** If a categorical feature has k possible values, you create k new binary features.

How it Works

Instead of one "Ear Shape" feature, we now create three new features:

1. Is_Pointy
2. Is_Floppy
3. Is_Oval

An example is "encoded" by placing a 1 in the column corresponding to its value and 0s in all other columns.

Example Transformation:

- An animal with "Pointy" ears becomes:
 - Is_Pointy: **1**
 - Is_Floppy: **0**
 - Is_Oval: **0**
- An animal with "Oval" ears becomes:
 - Is_Pointy: **0**
 - Is_Floppy: **0**
 - Is_Oval: **1**

This is called **"one-hot"** because for any given example, exactly *one* of these new features is "hot" (set to 1).

Benefits and Applications

- **For Decision Trees:** This transformation converts the complex categorical feature back into a set of simple binary features. Our decision tree learning algorithm (using information gain) can then use these new 0/1 features without any modification.
- **For Other Models:** One-hot encoding is a standard technique used to prepare categorical data for **any** model that requires numerical inputs, including **neural networks** and **logistic regression**.